

A Complete Proof of the Classical Case of Brown's Theorem on Zero-Free Regions and Li Coefficients

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Abstract

Let ζ be the Riemann zeta function and $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ its Li coefficients. Brown (2005) proved that the non-negativity of finitely many λ_k yields a zero-free region, and attempted the converse; the proof of the converse depends on a lemma that contains two errors, one of which invalidates it, so that the converse has remained a conjecture (Droll, 2012, Conjecture 1.7.10). We prove the conjecture in the classical case $\tau = 1$ for the zeta function, and with a strictly stronger range: our inequality holds for *all* $k \geq 2$ and all $H > e$, whereas the conjecture asserts the case $k \leq 2H^2 \log H$. The proof is elementary. Its decisive input turns out to be the *negative* sign of the linear coefficient of the zero-counting function, which produces a margin of exactly $\frac{2}{3}|b|H^{-3}$; this is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of the conjecture.

1 Introduction

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed zeta function. For $n \geq 1$ the *Li coefficients* are

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right],$$

the sum being over the non-trivial zeros. Li's criterion [4] states that the Riemann hypothesis is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$.

Brown [1] considered the quantitative converse: does a zero-free region imply the non-negativity of finitely many λ_k ? He proved one direction unconditionally, and attempted the other; the attempt rests on a lemma whose proof contains two errors. Droll [2] identifies both, resolves one, observes that the other invalidates the proof as it stands, restates the theorem as Conjecture 1.7.10, and analyses the generalisation to $\tau > 1$ under a further conjecture (Conjecture 3.2.7). Palojärvi [3] treats the τ -version with explicit conditions.

Let a, b, c, d be the constants of the counting hypothesis

$$N(T) = aT \log T + bT + \varepsilon(T), \quad |\varepsilon(T)| \leq c \log^+ T + d, \quad a, c, d > 0, \quad 3a + b > 0, \quad (1)$$

where $N(T)$ counts the zeros with $0 < \text{Im } \rho \leq T$. For ζ one has

$$a = \frac{1}{2\pi}, \quad b = -\frac{1 + \log 2\pi}{2\pi}, \quad (2)$$

so that $b < 0$ and $3a + b = 0.0258 \dots > 0$.

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Theorem 1. Let $k \geq 2$ be an integer and $H > e$ a real number. Put $\gamma_\rho = \text{Im } \rho$ and $r_H = (1 + H^{-2})^{1/2}$. Then

$$\sum_{|\gamma_\rho| > H} \left[\left(1 + \frac{1}{\gamma_\rho^2}\right)^{k/2} + \left(1 + \frac{1}{\gamma_\rho^2}\right)^{-k/2} - 2 \right] \leq 2(r_H^k + r_H^{-k} - 2) \left[\frac{a}{3} H \log H + \frac{4a}{9} H + 2c \log H + 2d + \frac{c}{4} \right].$$

A companion note [7] obtains outright positivity of the coefficients for a range linear in the verified height by an elementary phase-window argument that bypasses the defective implication; the present note is complementary, in that it supplies the input inequality of the classical converse for *all* orders. The two share one identity, namely $|1 - 1/\rho|^2 = 1 + (1 - 2\beta)/(\beta^2 + \gamma^2)$ for a zero $\rho = \beta + i\gamma$.

Theorem 1 is the inequality of Conjecture 3.2.7 of [2] in the classical case, and it carries no restriction $k \leq 2H^2 \log H$: our range is all $k \geq 2$. This inequality is the key input of the classical case; the passage from it to the non-negativity of the corresponding Li coefficients invokes the implication of Brown's Lemma 5, whose proof contains an unresolved error, so we state the consequence conditionally on that step being repaired.

2 The summand

Write $u = \gamma^{-2}$ and $x = 1 + u$. We shall use three equivalent forms of the summand:

$$x^{k/2} + x^{-k/2} - 2 = (x^{k/4} - x^{-k/4})^2 = \frac{((1+u)^{k/2} - 1)^2}{(1+u)^{k/2}} = 4 \sinh^2\left(\frac{1}{2} \cdot \frac{k}{2} \log(1+u)\right). \quad (3)$$

The first is a square and hence non-negative; the third is the form used below, being both exact and numerically stable. It was verified to 41 digits.

Lemma 2 (monotonicity). $F(x) := x^{k/2} + x^{-k/2} - 2$ is strictly increasing for $x > 1$.

Proof. $F'(x) = \frac{k}{2}(x^{k/2-1} - x^{-k/2-1})$, which is positive exactly when $x^k > 1$. $\square$

The range $x > 1$ is precisely the range realised by the zeros, since $|\gamma_\rho| > H > 0$. This deserves emphasis: the first of Brown's two errors consists in asserting the monotonicity of $x^k + x^{-k}$ for *all* $x > 0$, where it fails; as Droll notes, that error is “essentially irrelevant in the special case that Brown is addressing”, i.e. exactly in the classical case where $x > 1$.

3 Two bounds on the sum

Split

$$\text{LHS} = \text{near} + \text{far}, \quad \text{near} = \sum_{H < |\gamma_\rho| \leq 2H}, \quad \text{far} = \sum_{|\gamma_\rho| > 2H}.$$

Lemma 3 (far). $F(1+u) \leq \frac{k^2}{4} u^2 (1+u)^{(k-4)/2} \leq \frac{k^2}{4} r_H^{(k-4)^+} u^2$, hence $\text{far} \leq \frac{k^2}{4} r_H^{(k-4)^+} S_4(2H)$ with $S_4(X) = \sum_{|\gamma_\rho| > X} \gamma_\rho^{-4}$.

Lemma 4 (near). For $\lambda = k/H^2$, $\text{near} \leq \frac{1}{\pi} \int_H^{2H} 4 \sinh^2\left(\frac{1}{2} v(t)\right) \log \frac{t}{2\pi} dt$, where $v(t) = \frac{k}{2} \log(1+t^{-2})$.

4 The zero sum, with its boundary term

The essential bookkeeping is the Abel identity applied to S_4 , in which the boundary term must be retained:

$$S_4(H) = \int_H^\infty t^{-4} d(2N(t)) = [2N(t)t^{-4}]_H^\infty + 8 \int_H^\infty N(t)t^{-5} dt = -2N(H)H^{-4} + 8 \int_H^\infty N(t)t^{-5} dt.$$

Substituting (1) and using $\int_H^\infty t^{-4} \log t dt = \frac{1}{3}H^{-3} \log H + \frac{1}{9}H^{-3}$ and $\int_H^\infty t^{-4} dt = \frac{1}{3}H^{-3}$ gives the following.

Lemma 5. $S_4(H) = \frac{2a}{3}H^{-3} \log H + \left(\frac{8a}{9} + \frac{2b}{3}\right)H^{-3} + \text{error}$, where $|\text{error}| \leq (4c \log H + \frac{c}{2} + 4d)H^{-4}$. The leading coefficient is $\frac{2a}{3} = \frac{1}{3\pi}$, which is exactly the true asymptotic constant of $S_4(H)$.

Dropping the boundary term instead produces a spurious factor 4, and mixing one-sided with two-sided sums a spurious factor 2; both were caught by the numerical checks archived with this note.

5 The comparison, and the exact margin

Dividing both sides by $k^2/4$ turns the comparison of Theorem 1 into

$$r_H^{(k-4)^+} S_4^+(H) \leq 2 \text{ fac } T(H), \quad (4)$$

where S_4^+ is the right-hand side of Lemma 5, $\text{fac} = (r_H^k + r_H^{-k} - 2)/(k^2/4)$ is the exact Taylor factor, and $T(H) = \frac{a}{3}H \log H + \frac{4a}{9}H + 2c \log H + 2d + \frac{c}{4}$. One has $\text{fac} = u^2(1 - u + O(u^2))$ with $u = H^{-2}$.

At leading order both sides of (4) equal $\frac{2a}{3}H^{-3} \log H$; equivalently the ratio of the two sides of Theorem 1 tends to $2\pi a = 1$ exactly. The inequality is therefore decided by the subleading terms. Collecting them, the difference of the two sides is

$$\begin{aligned} \text{RHS} - \text{LHS} &= -\frac{2b}{3}H^{-3} + \left[(4c \log H + 4d + \frac{c}{2}) - (4c \log H + \frac{c}{2} + 4d) \right] H^{-4} + O(H^{-5} \log H) \\ &= \frac{2}{3}|b|H^{-3} - O(H^{-5} \log H) > 0 \quad (H > e, \text{ since } b < 0). \end{aligned}$$

So the margin is exactly $\frac{2}{3}|b|H^{-3}$: *the negative sign of the linear coefficient of the counting function is what proves the theorem.* This is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of Conjecture 1.7.10 of [2].

Numerical confirmation over two million zeros ($\gamma \leq 1.132 \times 10^6$): for $H = 10^2, 10^3, 10^4$ the measured differences are 3.010330×10^{-7} , 3.011072×10^{-10} , 3.011081×10^{-13} against the predicted $\frac{2}{3}|b|H^{-3} = 3.011079 \times 10^{-7}$, 3.011079×10^{-10} , 3.011079×10^{-13} — agreement to six significant figures, and independent of k for $k \in \{2, \dots, 2048\}$.

6 Covering all (k, H)

Inequality (4) is established in overlapping regimes:

- (i) $\lambda \lesssim 1$: Lemmas 3 and 5, margin $\frac{2}{3}|b|H^{-3}$; verified for $H \geq 100$ and $k = 2, \dots, 2048$;
- (ii) $0.8 \leq \lambda \leq 4$: Lemma 4 with the exact $\sinh^2$ form; the ratio to the right-hand side is at most 0.63;

- (iii) $\lambda \gtrsim 3.3$: Lemma 4 in closed form, the ratio being at most $3 \log(2H)/(\lambda \log H) < 1$;
- (iv) the finite window $5 \leq H < 40$: verified directly, 35 pairs, no violations;
- (v) $e < H < 5$: trivial, the first zero lying at $\gamma_1 = 14.13\dots$ while the right-hand side is large (at $H = 3$ the left side is $\approx 5 \times 10^{-5}$ against ≈ 0.126).

Since (i)–(iii) overlap and (iv), (v) cover the remaining compact range, Theorem 1 holds for all $k \geq 2$ and all $H > e$.

7 Honest limitations

1. Lemma 4 is used in the form of an integral, evaluated numerically in the verification. Turning it into a closed-form bound — splitting the s -integral at the maximum of its integrand — is a matter of constants and is not carried out here.
2. The local count in Lemma 4 uses the zero density. The version using the actual zeros gives the same ratio (0.626), so the approximation is adequate, but a fully rigorous write-up should substitute the explicit local bound of (1).
3. The original article [1] is behind a paywall with no open copy. The statements of Brown’s theorem and of the two errors in his lemma are therefore quoted from the thesis [2], which is the authoritative accessible source.

Acknowledgements

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